Assessment uncertainty for the shortcut MSE of ane.27.9aS

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Stock: Anchovy (*Engraulis encrasicolus*) in the southern part of Division 9.a (Gulf of Cadiz and southern part of Portugal), ane.27.9aS

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1 General approach

Management Strategy Evaluation (MSE) uses closed-loop simulations to represent population dynamics, the perception of stock status, and the implementation of management measures (Punt *et al.*, 2016).

In a full MSE, the assessment model can be run during each management cycle using data generated by the Operating Model (OM). In the *shortcut MSE* developed for Gulf of Cadiz anchovy (*Engraulis encrasicolus*, ane.27.9aS), Stock Synthesis 3 (SS3) is not refitted during the projection. Assessment uncertainty is represented by a statistical emulator that introduces error into the perception of the stock quantities used by the management procedure (Wiedenmann *et al.*, 2015; ICES, 2020).

The simulation cycle therefore distinguishes between the population state represented by the OM and the perceived state used by the management procedure:

Operating Model ⟶ assessment-error emulator ⟶ Management Procedure ⟶ management action ⟶ Operating Model.

Assessment uncertainty is kept separate from the uncertainty used to condition the OM. The 100 OMs represent alternative population conditions derived from historical conditioning, whereas *assessment error* represents error in the perception of those conditions during the projection.

The emulator was parameterized from historical revisions of the SS3 assessment for ane.27.9aS. These revisions were used to estimate bias, variability, correlation among variables, and temporal dependence of assessment error.

The *assessment error* was parameterized jointly for spawning-stock biomass (SSB), fishing mortality (F), and recruitment (R). All three variables are retained in the assessment-error object. The reference MP documented in WD06 uses the SSB and F multipliers, whereas recruitment assessment error remains disabled.

This WD05 documents the parameterization and validation of the assessment error used in the *shortcut MSE*. WD06 documents its incorporation into the perceived state used by the MP, and subsequent WDs evaluate its consequences for management-procedure performance.

2 Methods

2.1 Historical characterization of *assessment error*

2.1.1 SS3 assessments used

The *assessment error* was parameterized from a sequence of SS3 assessments with terminal years from 2004 to 2024. The assessments used the same model configuration and differed in terminal year.

For each run, convergence was checked using the maximum gradient reported by SS3. Runs satisfying the following criterion were included:

$$\text{max gradient} < 0.001.$$

Positive and finite estimates of spawning-stock biomass (SSB), fishing mortality (F), and recruitment (R) were used to calculate *assessment error*.

The sequence provided 20 annual updates for the period 2005–2024.

2.1.2 Annual revisions

For each variable X and update year t, the estimate for year $t-1$ obtained from the assessment with terminal year $t-1$ was compared with the estimate for the same year obtained from the assessment with terminal year t:

$$e_{X,t} = \log\left(\frac{\hat{X}_{t-1}^{(t-1)}}{\hat{X}_{t-1}^{(t)}}\right),$$

where $\hat{X}_{t-1}^{(t-1)}$ is the estimate available before incorporating the new year of information and $\hat{X}_{t-1}^{(t)}$ is the revised estimate after incorporating it.

The relative revision was also calculated:

$$r_{X,t} = \frac{\hat{X}_{t-1}^{(t-1)}}{\hat{X}_{t-1}^{(t)}} - 1.$$

With this definition, positive values indicate that the previous estimate was higher than the revised estimate, whereas negative values indicate that it was lower.

The revisions were used as a proxy for assessment error. They do not represent differences from the true stock state, which is unobserved. Their use allows characterization of changes in estimates resulting from the incorporation of additional information into the assessment (Mohn, 1999; Hurtado-Ferro *et al.*, 2015).

2.1.3 Error parameterization

For each update, the log revisions of SSB, F, and R were arranged in the vector

$$\mathbf{e}_t = \begin{pmatrix} e_{\mathrm{SSB},t} \\ e_{F,t} \\ e_{R,t} \end{pmatrix}.$$

From the $n = 20$ vectors, the mean vector was estimated as

$$\boldsymbol{\mu} = \frac{1}{n}\sum_{t=1}^{n} \mathbf{e}_t$$

and the variance–covariance matrix as

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{e}_t).$$

The empirical mean was retained in the parameterization and the following correction was not applied:

$$-\frac{\sigma_X^2}{2}.$$

Therefore, the multipliers generated from the log revisions retain the mean bias estimated from the historical revisions and are not adjusted to impose an expectation equal to one on the multiplicative scale.

The full $\boldsymbol{\Sigma}$ matrix was used, retaining the estimated covariation among the revisions of SSB, F, and R.

The stability of the relationship between SSB and F revisions was examined using a *leave-one-year-out* analysis. For each update year, the corresponding vector was removed and Pearson and Spearman correlations between the two variables were recalculated.

2.1.4 Temporal structure

Temporal dependence of the revisions was evaluated separately for SSB, F, and R using

$$e_{X,t} = \alpha_X + \phi_X e_{X,t-1} + \varepsilon_{X,t}.$$

For each variable, the first-order coefficient ϕ_X and its p-value were estimated. These results were used to define the temporal dependence of assessment error in the reference scenario.

2.2 *Assessment-error* emulator

In the *shortcut MSE*, SS3 is not refitted during each annual management cycle. Assessment uncertainty is represented by a statistical emulator parameterized from the historical revisions described in Section 2.1.

For each year t, the reference model generates

$$\mathbf{e}_t \overset{\text{iid}}{\sim} \mathcal{N}_3\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}\right).$$

Independence is assumed among years, whereas dependence among SSB, F, and R within each year is determined by $\boldsymbol{\Sigma}$.

For each variable X, the error is transformed to the multiplicative scale as

$$M_{X,t} = \exp(e_{X,t}).$$

During the *closed-loop* simulation, perceived quantities are obtained as

$$X_t^{\text{perc}} = X_t^{\text{true}} M_{X,t},$$

where X_t^{true} is the state generated by the OM and X_t^{perc} is the perceived state used by the management procedure.

This WD parameterizes and validates the *assessment-error* generator. The application of the multipliers to the perceived state and their propagation to management advice are subsequently checked with the Management Procedure $F_{cap} + B_{esc}$.

2.3 Emulator validation

2.3.1 Internal validation

The implementation of the multivariate model was checked using 10,000 independent realizations:

$$\mathbf{e}_s \sim \mathcal{N}_3\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}\right), \qquad s = 1, \ldots, 10000.$$

A fixed seed (`set.seed(123)`) was used. For each variable, the mean and standard deviation of the realizations were calculated together with the correlation matrix among variables. These statistics were compared with the $\boldsymbol{\mu}$ vector, the standard deviations derived from $\boldsymbol{\Sigma}$, and the correlation matrix used to parameterize the emulator.

This check evaluates whether the generator reproduces the specified moments and correlation structure. It does not evaluate the assumption of multivariate normality of the historical revisions.

Five temporal sequences were additionally generated as a graphical diagnostic of the reference process. These sequences are not part of the 1,000 realizations subsequently used in the MSE.

2.3.2 Validation using conventional retrospectives

Validation using information not used in the parameterization was conducted with conventional SS3 retrospectives corresponding to *peels* -1 to -5. These runs were not used to estimate $\boldsymbol{\mu}$ or $\boldsymbol{\Sigma}$.

For each variable X and *peel* k, the observed retrospective error was calculated as

$$E_{X,k}^{\text{obs}} = \log\left(\frac{\hat{X}_{Y-k}^{(-k)}}{\hat{X}_{Y-k}^{(0)}}\right),$$

where Y is the terminal year of the full assessment, $\hat{X}_{Y-k}^{(-k)}$ is the terminal estimate from retrospective *peel* k, and $\hat{X}_{Y-k}^{(0)}$ is the estimate for the same year in the full assessment.

Because a k-year *peel* represents the accumulation of k annual updates, the simulated retrospective error was calculated as

$$E_{X,k}^{\text{sim}} = \sum_{j=1}^{k} e_{X,j}.$$

Ten thousand sequences of five annual errors were generated. For each variable and *peel*, the median and 5th and 95th percentiles of $E_{X,k}^{\text{sim}}$ were calculated, and the observed retrospective error was compared with this distribution.

As a second diagnostic, Mohn's rho was calculated (Mohn, 1999):

$$\rho_X = \frac{1}{K}\sum_{k=1}^{K}\left[\frac{\hat{X}_{Y-k}^{(-k)}}{\hat{X}_{Y-k}^{(0)}} - 1\right],$$

con $K = 5$.

The same statistic was calculated for each simulated sequence. For each variable, the median and 5th and 95th percentiles of the simulated distribution of ρ_X were obtained and compared with the observed value. Mohn's rho was used to characterize the retrospective pattern and not as an estimate of error relative to the true population state (Mohn, 1999; Hurtado-Ferro *et al.*, 2015).

As a third diagnostic, persistence in the sign of retrospective errors was evaluated. For each variable, the number of *peels* with positive and negative errors was counted. The same calculation was performed for each simulated sequence, and the frequency with which the observed directional pattern appeared in the simulations was estimated.

2.4 Generation of realizations used in the MSE

Once the emulator had been parameterized and validated, the *assessment-error* realizations subsequently used in the MSE were generated.

For each year t and iteration i,

$$\mathbf{e}_{t,i} \overset{\text{iid}}{\sim} \mathcal{N}_3\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}\right),$$

and the multipliers were calculated as

$$\mathbf{M}_{t,i} = \exp(\mathbf{e}_{t,i}).$$

Generation used a fixed seed (`set.seed(123)`) and produced 1,000 realizations for assessment years 2024–2053. The resulting object has dimensions

$$30 \text{ años} \times 1000 \text{ iteraciones} \times 3 \text{ variables}.$$

Before saving the object, it was checked that its dimensions matched the specified configuration, that the variables were SSB, F, and R, and that all multipliers were finite and positive.

As a check of the final generation, the multipliers were transformed back to the logarithmic scale and their means, standard deviations, and correlations were calculated. These statistics were compared with $\boldsymbol{\mu}$, the standard deviations, and the correlation matrix used to parameterize the emulator.

The subsequent application of the multipliers is defined as

$$X_{t,i}^{\text{perc}} = X_{t,i}^{\text{true}} M_{X,t,i}.$$

This transformation is neither applied nor validated in this WD. Its implementation on the perceived state used by the Management Procedure is documented in WD06.

3 Results

3.1 Historical revisions and parameterization

Twenty annual revision vectors were obtained for SSB, F, and recruitment, corresponding to SS3 assessment updates between 2005 and 2024. The trajectories of the assessments used to calculate the revisions are shown in Figure 1.

The mean log revisions were 0.025 para SSB, -0.041 para F y 0.082 for recruitment. The standard deviations were 0.191, 0.284 y 0.310, respectively (Table 1). The correlations among log revisions were -0.979 para SSB–F, 0.646 para SSB–R y -0.722 para F–R (Tabla 1). In the leave-one-year-out analysis, the SSB–F correlation ranged from -0.981 y -0.974 for Pearson, and from -0.995 y -0.989 for Spearman.

The annual variation in log revisions is shown in Figure 2.

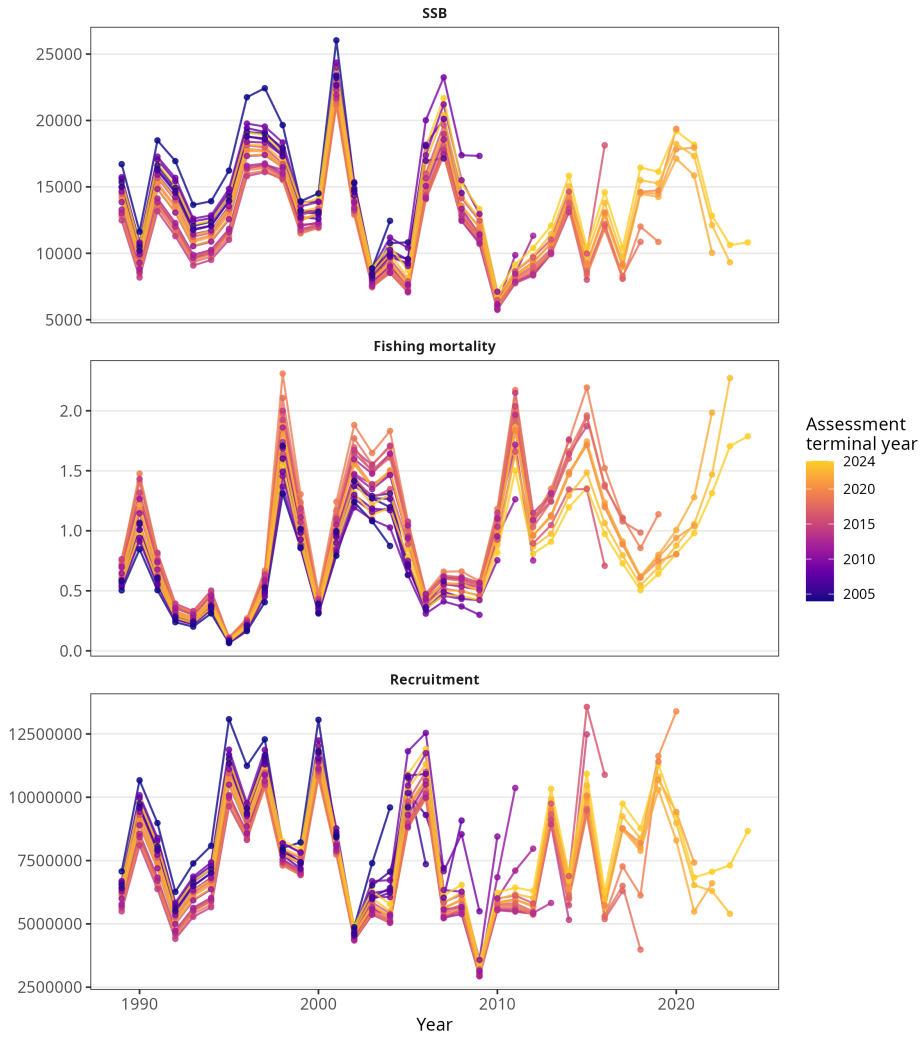


Figure 1: Trajectories of the SS3 assessments used to calculate annual revisions. Each line corresponds to an assessment with a terminal year between 2004 and 2024.

Table 1: Parameters estimated from the 20 annual SS3 revisions corresponding to 2005–2024.

Quantity	Mean log-error	SD log-error	SSB	F	R
SSB	0.025	0.191	1.000	-0.979	0.646
F	-0.041	0.284	-0.979	1.000	-0.722
R	0.082	0.310	0.646	-0.722	1.000

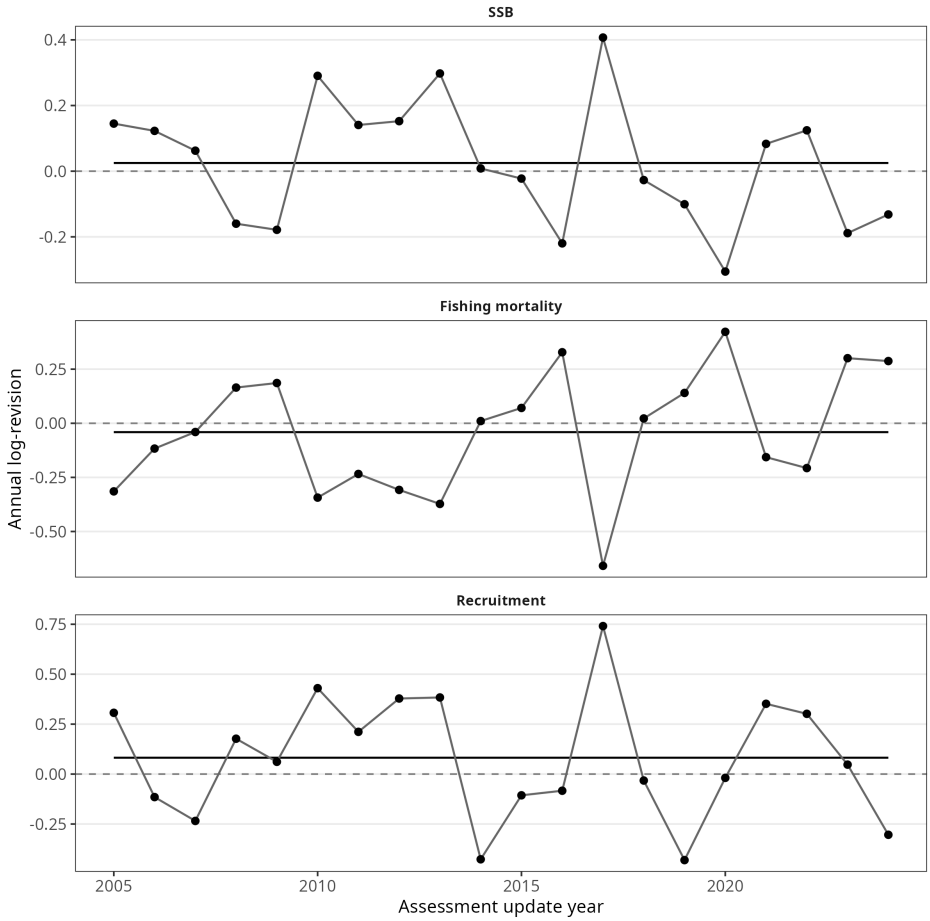


Figure 2: Annual log revisions of SSB, fishing mortality, and recruitment corresponding to the 2005–2024 updates. The dashed line represents zero error and the solid line the mean log revision for each variable.

3.2 Temporal structure

The first-order temporal-dependence coefficients were $\phi = 0.022$ para SSB, $\phi = 0.057$ para F y $\phi = 0.056$ for recruitment. The p-values were 0.930, 0.817 y 0.825, respectively (Table 2).

The first-order temporal-dependence coefficients were $\phi = 0.022$ para SSB, $\phi = 0.057$ para F y $\phi = 0.056$ para reclutamiento, con valores de p de 0.930, 0.817 y 0.825, respectively (Table 2). Con estos resultados, el modelo de referencia se implementó con $\phi = 0$ para las tres variables. La covariación contemporánea entre SSB, F y reclutamiento se mantuvo mediante la matriz de varianzas–covarianzas $\mathbf{\Sigma}$.

Table 2: First-order temporal-dependence coefficients estimated for annual revisions of SSB, fishing mortality, and recruitment.

variable	phi	p_value
SSB	0.022	0.930
F	0.057	0.817
R	0.056	0.825

3.3 Internal validation

In the 10,000 realizations used to check implementation of the emulator, the simulated means were 0.025, -0.041 y 0.080 for SSB, F, and recruitment, respectively. The values used in the parameterization were 0.025, -0.041 y 0.082.

The simulated standard deviations were 0.191, 0.283 y 0.311, compared with parameterization values of 0.191, 0.284 y 0.310, respectivamente.

The simulated correlations were -0.979 para SSB–F, 0.645 para SSB–R y -0.721 para F–R. The values used in the parameterization were -0.979, 0.646 y -0.722, respectively (Table 3).

Table 3: Comparison between the parameters used by the emulator and statistics calculated from 10,000 multivariate realizations.

Variable	Target mean	Simulated mean	Target SD	Simulated SD	Target cor. with SSB	Simulated cor. with SSB
SSB	0.025	0.025	0.191	0.191	1.000	1.000
F	-0.041	-0.041	0.284	0.283	-0.979	-0.979
R	0.082	0.080	0.310	0.311	0.646	0.645

The log-error distributions generated by the emulator and the historical log revisions used for its parameterization are shown in Figure 3.

Five sequences generated by the reference model are shown in Figure 4. These sequences were used only as a graphical diagnostic and are not part of the 1,000 realizations used in the MSE.

3.4 Validation using conventional retrospectives

The observed retrospective errors for *peels* -1 to -5 were compared with the distributions obtained from 10,000 simulated sequences. All 15 observed errors, corresponding to five *peels* and three variables, fell within the 5th and 95th percentiles of the simulated distributions (Table 4).

For SSB, observed errors ranged from -0.396 to 0.008; for F, from -0.007 to 0.574; and for recruitment, from -0.304 to 0.363.

Observed Mohn's rho was -0.134 for SSB, 0.335 for F, and 0.041 for recruitment. All three values fell within the 5th and 95th percentiles of the corresponding simulated distributions (Table 5).

Four of the five peels showed positive errors for F, three for recruitment, and one for SSB. The frequencies of these patterns in the 10,000 simulated sequences were 0.281, 0.702, and 0.288, respectively (Table 6).

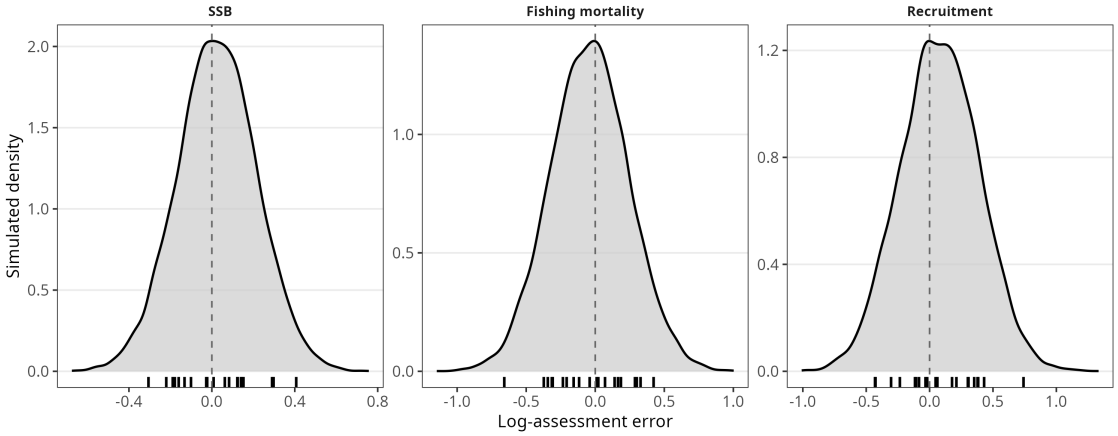


Figure 3: Distribuciones de log-error generadas por el emulador y log-historical revisions utilizadas para su parametrización. Las curvas corresponden a las realizaciones simuladas y las marcas sobre el eje horizontal a las historical revisions.

Table 4: Observed retrospective error and simulated distribution for peels -1 to -5.

Peel	Year	Variable	Observed	Median	p05	p95	Within 90%
-1	2023	F	0.287	-0.042	-0.511	0.434	TRUE
-2	2022	F	0.413	-0.081	-0.753	0.573	TRUE
-3	2021	F	0.060	-0.127	-0.939	0.685	TRUE
-4	2020	F	-0.007	-0.167	-1.086	0.781	TRUE
-5	2019	F	0.574	-0.209	-1.242	0.860	TRUE
-1	2023	R	-0.304	0.083	-0.428	0.600	TRUE
-2	2022	R	-0.067	0.168	-0.544	0.898	TRUE
-3	2021	R	0.082	0.250	-0.652	1.142	TRUE
-4	2020	R	0.363	0.340	-0.693	1.352	TRUE
-5	2019	R	0.010	0.420	-0.731	1.558	TRUE
-1	2023	SSB	-0.132	0.027	-0.289	0.340	TRUE
-2	2022	SSB	-0.246	0.050	-0.393	0.504	TRUE
-3	2021	SSB	-0.012	0.078	-0.467	0.623	TRUE
-4	2020	SSB	0.008	0.100	-0.537	0.728	TRUE
-5	2019	SSB	-0.396	0.130	-0.584	0.819	TRUE

Table 5: Observed Mohn's rho and distribution obtained from 10,000 simulated sequences.

Variable	Observed rho	Median	p05	p95	Within 90%
F	0.335	-0.099	-0.535	0.848	TRUE
R	0.041	0.330	-0.376	2.021	TRUE
SSB	-0.134	0.094	-0.308	0.761	TRUE

Table 6: Directional persistence of retrospective errors and corresponding frequency in simulated sequences.

Variable	Positive	Negative	Frequency
F	4	1	0.281
R	3	2	0.702
SSB	1	4	0.288

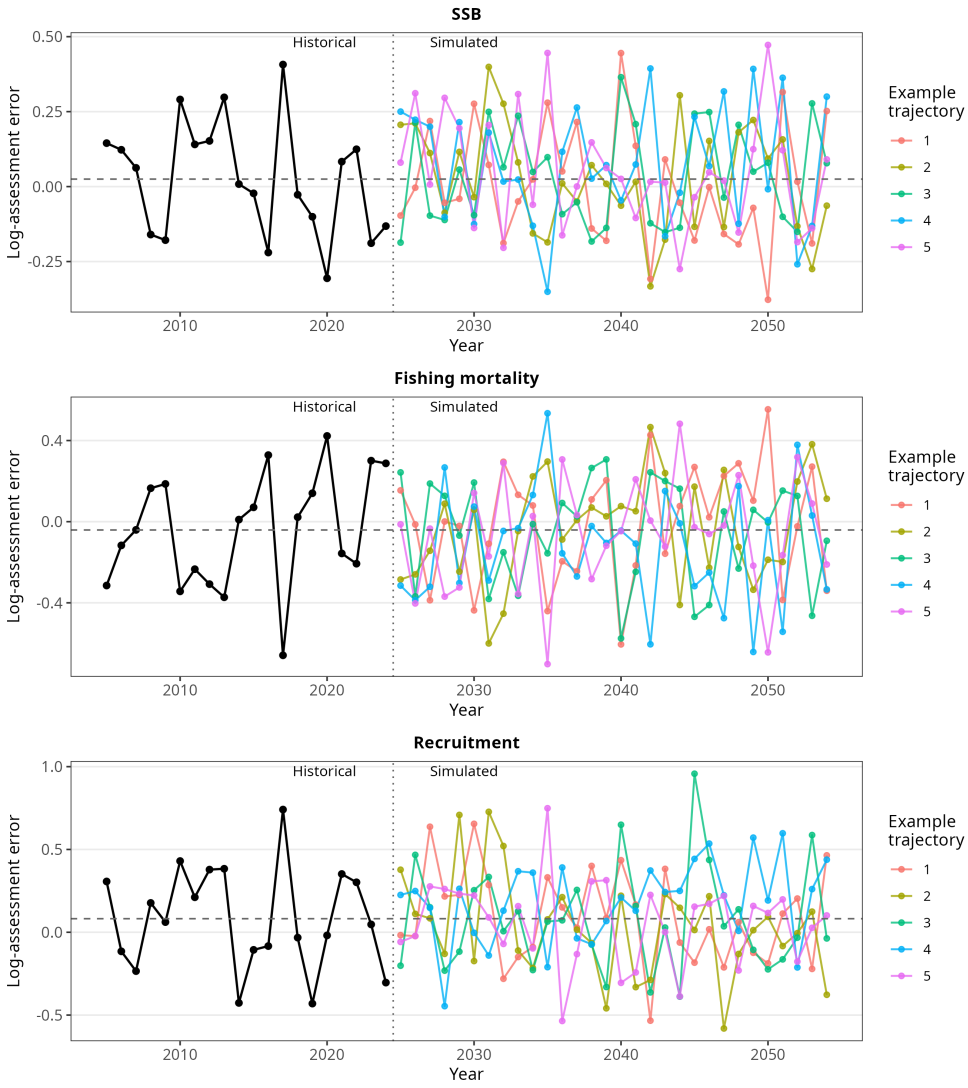


Figure 4: Cinco secuencias de log-error generadas por el modelo de referencia. Las líneas negras corresponden a las historical revisions y las líneas de color a las realizaciones simuladas.

The accumulated errors by peel, Mohn's rho, and directional persistence are shown together in Figure 5.

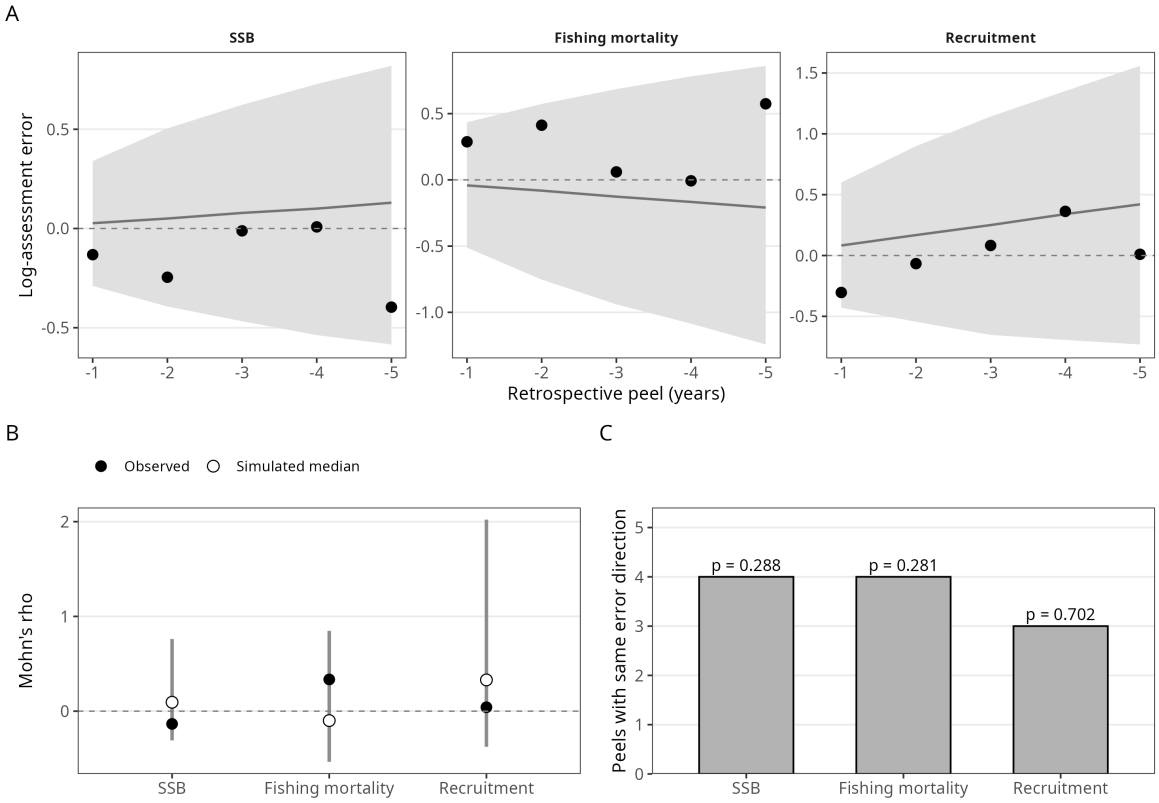


Figure 5: Validation using conventional SS3 retrospectives. (A) Observed retrospective error by peel and median and 5th and 95th percentiles of simulations. (B) Observed Mohn's rho and simulated distribution. (C) Observed directional persistence and corresponding frequency under the simulated process.

3.5 Realizations used in the MSE

A total of 1,000 multivariate *assessment-error* realizations were generated for assessment years 2024–2053. The final object contains 30 years, 1,000 iterations, and three variables: SSB, F, and recruitment.

Checks of the final object confirmed the specified dimensions, the presence of all three variables, and the absence of non-finite or non-positive multipliers.

The object was stored as

`data/Rdata/assessment_error_reference_1000iter.rds`

and constitutes the *assessment-error* input subsequently used in the MSE *closed-loop* simulations.

4 Reproducibility

The *assessment-error* component is generated by the script `14_build_assessment_error.R`. The script uses moving SS3 retrospectives with terminal years 2004–2024 to calculate annual revisions and parameterize the emulator. Conventional retrospectives with *peels* 0 to -5 are used for validation.

The script calculates historical revisions of SSB, F, and recruitment, estimates the parameters of the multivariate model, performs validation diagnostics, and generates the 1,000 *assessment-error* realizations subsequently used in the MSE.

The computational sequence is:

SS3 retrospectives
↓
historical revisions
↓
emulator parameterization and validation
↓
`assessment_error_reference_1000iter.rds`

The final object is stored in `data/Rdata/assessment_error_reference_1000iter.rds`. The tables used in this WD are stored in `outputs/assessment_error/tables/` and the figures in `outputs/assessment_error/figures/`.

The R Markdown does not rerun SS3 or regenerate the *assessment-error* realizations. Numerical results and figures are read from the outputs generated by `14_build_assessment_error.R`, maintaining traceability among code, outputs, and reported results.

5 Conclusions

- Las historical revisions de SS3 mostraron diferencias en magnitud y dirección entre SSB, mortalidad por pesca y reclutamiento, con desviaciones estándar de 0.191, 0.284 y 0.310 en escala logarítmica, respectivamente.
- SSB and fishing-mortality revisions showed a negative correlation of -0.979, which was maintained in the *leave-one-year-out* analysis. This relationship was incorporated into the emulator through the variance–covariance matrix.
- The first-order temporal-dependence coefficients were próximos a cero para SSB, mortalidad por pesca y reclutamiento, sin evidencia estadística de dependencia temporal. El modelo de referencia se implementó con errores independientes entre años y mantuvo la dependencia entre variables dentro de cada año.
- Emulator simulations reproduced the means, standard deviations, and correlations used in its parameterization.
- Observed errors in the conventional retrospectives for *peels* -1 to -5 and Mohn's rho values fell within the 5th and 95th percentiles of the simulated distributions for all three variables.
- The results support use of the emulator as a representation of *assessment error* in the *shortcut MSE*. The 1,000 realizations generated for 2024–2053 provide the multipliers that will subsequently be applied to the state perceived by the Management Procedure.

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